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Oxyfluoride tellurite glasses doped by erbium: thermal analysis, structural organization and spectral properties

V. Nazabal [a,b,*], S. Todoroki [a], A. Nukui [a], T. Matsumoto [a], S. Suehara [a], T. Hondo [c], T. Araki [c], S. Inoue [a], C. Rivero [d], T. Cardinal [d]

[a] *National Institute for Materials Sciences, Advanced Materials Laboratory, Tsukuba, Ibaraki 305-0044, Japan*
[b] *Laboratoire Verres et Céramiques, Institut de Chimie de Rennes, Université de Rennes 1, UMR 6512, Bât. 10A, Campus de Beaulieu CS 74205, 35042 Rennes cedex, France*
[c] *Department of Materials Science and Technology, Science University of Tokyo, Noda, Chiba 278-8510, Japan*
[d] *Institut de Chimie de la Matière Condensée de Bordeaux (ICMCB), CNRS UPR 9048, 33608 Pessac cedex, France*

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Abstract

The effects of fluorine substitution in zinc tellurite glass system doped with rare earths on the spectral properties of the Er^{3+} ions are investigated. Differential thermal analysis, vibrational spectroscopies studies and in situ high-temperature X-ray diffraction measurements have been considered in term of fluorine influence. As a function of composition, we have principally measured optical absorption, spontaneous emission, and lifetime measurements. Judd–Ofelt intensity parameters of Er^{3+} in these host glasses were determined and used to calculate radiative transition rates and lifetimes. The bulk composition variation and addition of fluoride compounds in tellurite glasses result among others, in broad emission spectra, improved emission lifetime and difference in relative band intensities compared to pure oxide glass.

1. Introduction

The incorporation of rare-earth ions into a variety of glasses has been a key, in the last decade, to the development of optical devices based on active glass materials, such as infrared (IR) lasers, IR-to-visible up-converters, fiber and waveguide amplifiers for optical transmission network. The latest evolution of telecommunication networks maintains attention on this research subject: wavelength division multiplexing systems require amplifiers materials having the broadest possible emission spectrum and an increasing demand is perceived for compact optical amplifiers in order to supply low cost optical devices. Glasses based on tellurium oxide doped by Er^{3+} ions can provide a broadband amplification around 1530 nm ($^4I_{13/2} \rightarrow {^4I_{15/2}}$ transition). Their low phonon

* Corresponding author. Address: Laboratoire Verres et Céramiques, Institut de Chimie de Rennes, Université de Rennes 1, UMR 6512, Bât. 10A, Campus de Beaulieu CS 74205, 35042 Rennes cedex, France. Tel.: +33-2 23 23 57 48; fax: +33-2 23 23 56 11.
E-mail address: nazabal@univ-rennes1.fr (V. Nazabal).

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energy can reduce the non-radiative transition probability of the doped rare-earth ions compared to silicate, phosphate or borate glasses [1]. Furthermore, tellurite glass has a large refractive index, which has an influence on the emission cross-section of rare-earth ions [2]. Wide infrared transmittance, good chemical durability are the other attractive characteristics recently taken into account for the development of an Er^{3+}-doped fiber amplifier based on tellurite glasses (EDTFA) [3]. Moreover, the modification of the rare-earth spectral properties by addition of fluorine within tellurite glasses can be promising to obtain a broad amplifier bandwidth with an intrinsically flat gain required in the development of optical transmission network [4]. The creation of a local bonding environment for the rare earth elements that are a combination of oxide and fluoride like sites should modify the characteristic of their spectral properties. Furthermore, the presence of hydroxyl within the glass matrix has a great influence on quenching effects on the excited levels of Er^{3+} concerned with the radiative emission. A decrease of the hydroxyl concentration can be expected in these oxyfluoride glasses and consequently, the damaging effect induced by energy migration to impurities like hydroxyl groups should be limited [5]. In this article, we report and discuss the effect of host composition in particular the addition of increased amounts of fluorine to the oxide glasses in term of optical absorption, emission spectra, up-conversion luminescence and lifetime of Er^{3+} doped in oxyfluoride zinc tellurite glasses. In parallel, we investigated the modification of the host glass network induced by fluoride compound introduction with Raman and IR spectroscopy studies and in situ high-temperature X-ray diffraction measurements in order to better manage the influence of composition on optical properties.

2. Experimental procedure

Commercial reagent grade powders with purity of 99.99% (TeO_2, ZnO, ZnF_2, Er_2O_3, GeO_2, ErF_3) were used to prepare the glasses (6 g batch). Doped oxide and oxyfluoride tellurite glasses were obtained by addition to the batch of the desired rare-earth oxide or fluoride. The appropriate mixtures introduced in platinum crucible and covered with a platinum plate to avoid vaporization losses were melted in an electric furnace at 650–800 °C for 20 min depends on composition. The glasses prepared by this conventional melting and quenching techniques were annealed for 3 h at 30 °C below their glass transition temperatures in order to reduce strains. All samples about 2 mm thick were polished to reduce light scattering in optical measurements. The compositions of glasses were analyzed with AES–ICP (atomic emission spectroscopy–inductively coupled plasma), and with XPS (X-ray photoelectron spectroscopy) to confirm the fluorine concentration obtained with AES–ICP (Table 1). The thermal properties (glass transition (T_g), onset crystallization (T_x), crystallization peak (T_c) temperatures) have accurately been analyzed by differential thermal analysis at a heating rate of 10 °C/min in the temperature range of 25–700 °C with α-Al_2O_3 used as standard reference and found reproducible within ±5 °C. The densities were measured according to the Archimedes principle using calibrated CCl_4 as liquid

Table 1
Compositions of the studied oxide and oxyfluoride tellurite glasses and the ratio zinc/tellurium and fluorine/oxygen

Glass matrix	Composition (mol%)				
	TeO_2	ZnO	ZnF_2	$R_{Zn/Te}$	$R_{F/O}$
ETZ	75.1 ± 0.4	24.9 ± 0.1		0.33	0
ETZF-1	74.7 ± 0.4	18.3 ± 0.1	7.1 ± 0.1	0.34	0.08
ETZF-2	74.6 ± 0.4	8.8 ± 0.1	16.6 ± 0.1	0.34	0.21
ETZF-3	69.0 ± 0.3	9.3 ± 0.1	21.7 ± 0.1	0.45	0.29
ETZF-4	60.5 ± 0.3	12.0 ± 0.1	27.5 ± 0.1	0.65	0.41
ETZF-5	46.6 ± 0.2	18.2 ± 0.1	35.2 ± 0.2	1.15	0.63

Table 2
Values of glass transition (T_g), crystallization onset (T_x), peak crystallization (T_c) temperatures, volumetric weight (ρ) and cut-off wavelengths (λ_c)

Matrix	T_g ±5 °C	T_x ±5 °C	T_c ±5 °C	$T_x - T_g$ (°C)	$\rho \pm 0.01$ g/cm^3	λ_c (nm)
$75TeO_2$–25ZnO	315	404	420	89	5.53	357
$75TeO_2$–18ZnO–$7ZnF_2$	278	398	417	120	5.49	343
$75TeO_2$–9ZnO–$16ZnF_2$	261	380	398	119	5.45	331
$69TeO_2$–9ZnO–$22ZnF_2$	258	378	395	120	5.43	322
$61TeO_2$–12ZnO–$27ZnF_2$	257	374	389	117	5.37	311
$47TeO_2$–18ZnO–$35ZnF_2$	256	368	387	112	5.33	305

medium. The thermal properties and densities are listed in Table 2.

Raman scattering spectra were obtained using a FT-IR/Raman spectrophotometer. The 1064 nm emission line of a diode pumped Nd^{3+}:YAG laser was used for excitation with incident power around 200 mW. A Seeman–Bohlin geometry was adopted for the in situ high temperature X-ray diffractometry experiment. The choice of this X-ray optical system has been made due to its ability to produce rapid measurements allowing us to study intermediate phases and to attempt a direct comparison with DTA measurements made under similar conditions. The structural changes in the studied sample were observed using a high-temperature X-ray diffractometer equipped with a furnace and a position sensitive detector (PSD). A heating rate of 10 °C/min was adopted and the temperatures reported in this article with an accuracy about ±5 °C. X-ray diffraction patterns at each temperature step were obtained over a wide range of diffraction angles (20–130°), for a short measurement time of 60 s.

The visible and near-infrared absorption spectra were measured at room temperature using a UV–visible–nIR spectrophotometer over a spectral range of 250–3200 nm and were normalized to an 1mm thickness for all samples. Transmittance spectra were recorded using a Fourier transform infrared (FT-IR) spectrophotometer with 4 cm^{-1} resolution. The linear refractive indices were determined by the Brewster angle reflection method at 1064 nm (Nd^{3+}:YAG laser). Emission spectra for the 1.5 μm band of Er^{3+} were measured with a 974 nm pumping from a cw tunable near-infrared Ti–sapphire laser capable of tunable laser action over a broad range of near infrared wavelengths and also used for the up-conversion excitation. Fluorescence signals were coupled into a perpendicular oriented fiber recorded by a scanning spectrometer equipped with an InGaAs detector (900–1600 nm) or Si CCD (350–1050 nm). The spontaneous emission rates of $^4I_{13/2}$ and $^4I_{11/2}$ erbium levels were measured using a laser diode emitting at 975 nm and a high speed cooled germanium detector.

3. Results

3.1. Glass elaboration, thermal properties and densities

The analyzed composition of obtained oxyfluoride and oxide zinc tellurite glasses are closed to that of the initial batches; indeed, the ratio Te/Zn are comparable within 0.5–1% to started concentration. Within the respect of analyzed compositions, the designation of glasses compositions takes into account fluorine loss in order to keep things clear (Table 1). Furthermore, these glasses show a relatively high chemical durability and a low tendency to devitrification. They appeared to be of good optical quality, with no visual evidence of devitrification. These oxyfluoride tellurite glasses can be melted over an extensive compositional region in which rare-earth metals, such as erbium, are readily soluble. More accurately, these glasses were elaborated in the range of 75–47 mol% of TeO_2. The substitution of ZnF_2 to ZnO widely increases the glass forming range which is typically included in the range of 80–70 mol% at cooling rates of 1 K/min [6]. The typical DTA trace of the $0.75TeO_2$–$0.25ZnO$ (ETZ) glass is presented in comparison to other compositions with a raising proportion of fluorine from ETZF-1 to ETZF-5 (Fig. 1). The progressive insertion of fluorine with a gradual introduction of ($ZnO + ZnF_2$) in higher proportion affects the glass transition temperature following a same continuous tendency of decreasing, from 315 to 256 °C. The crystallization temperature is also influenced by the introduction of fluorine with a peak position shift. This moves toward lower temperature range involving an intermediated state well illustrated on the DTA trace of the ETZF-2 glass in which the crystallization peak shape is enlarged and reduced in magnitude. The melting temperature is also modified by the presence of fluorine; an endothermic peak around 509 °C is induced by substitution of ZnF_2 to ZnO present at lower temperature than oxide glass (625 °C) for ETZF-1 and 2 but insignificant for ETZF-5. The difference $T_x - T_g$ slightly increases with the introduction of fluorine and is large enough – greater than 110 °C – to obtain stable glasses (Table 2).

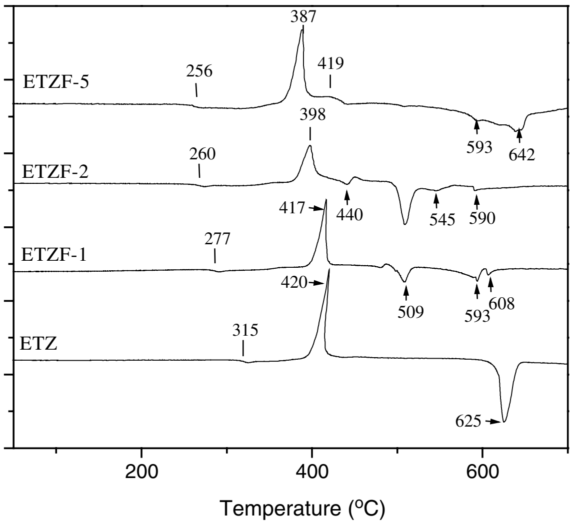


Fig. 1. DTA curves for the zinc tellurite oxide and oxyfluoride glasses $0.75TeO_2$–$(0.25 - x)ZnO$–$xZnF_2$ (with $x = 0, 7, 16$) and $47TeO_2$–$18ZnO$–$35ZnF_2$ (heating rate 10 °C/min) in the range temperature between room temperature to 700 °C.

3.2. Structure of the host-glass

The Raman spectra of oxyfluoride tellurite glasses studied are reported in Fig. 2 and present similar shape with relative oxide glasses with three main bands around 700–800, 660–670, 410–420 cm^{-1}. Like oxide glasses with similar composition, the Raman spectra of oxyfluoride glass samples show band intensity around 735–750 cm^{-1} a little higher than the band around 660 cm^{-1}. The increase of the $[ZnF_2 + ZnO]/[TeO_2]$ ratio results in a significant modification of the Raman spectra shape. Moreover, a shoulder at higher wavenumber (around 775 cm^{-1}) is observed for growing fluorine concentration. In parallel, the intensity of the band located at 420 cm^{-1} is affected.

Crystallization experiments versus temperature have been performed in order to examine the effect of fluorine addition on crystallization behaviors of oxyfluoride tellurite glasses. The XRD diagrams show a similar feature for the three first samples $75TeO_2$–$25ZnO$ (ETZ), $75TeO_2$–$18ZnO$–$7ZnF_2$ (ETZF-1), $75TeO_2$–$9ZnO$–$16ZnF_2$ (ETZF-2) whereas $47TeO_2$–$18ZnO$–$35ZnF_2$ (ETZF-5) sample present a divergent behavior. All the sample present a broad and high background characteristic of amorphous compounds and a mixture of several crystalline phases, particularly ETZF-5, in which the identification of distinct phases is more

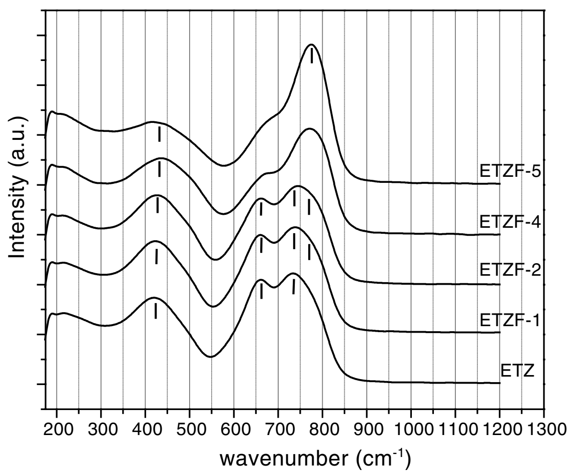
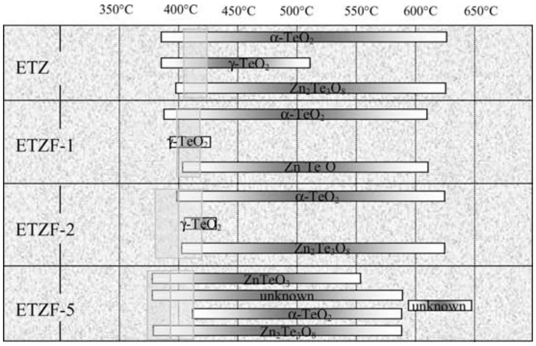


Fig. 2. Raman scattering spectra for a series of studied zinc tellurite oxide and oxyfluoride tellurite glasses.

Table 3
Representation of the crystallized phases from oxide and oxyfluoride tellurite glasses versus temperature



complex. For the sample ETZ, ETZF-1, ETZF-2, the crystalline phases in presence are clearly γ-TeO_2, α-TeO_2 paratellurite and $Zn_2Te_3O_8$ (Table 3 and Fig. 3). An oxyfluoride tellurite glass sample ETZF-1 submitted to a heat treatment during 30 min at 400 °C presents a limited crystallization localized on the surface sample and three additional narrower peaks at 394, 590 and 646 cm^{-1} appear on the Raman spectrum (Fig. 4). These peaks reveal the presence of α-TeO_2 crystalline phase within the glass [7] in agreement with the X-ray data versus temperature in which the formation of α-TeO_2 dominate the crystallization process at this temperature range. A γ-TeO_2 crystalline phase is clearly present in XRD pattern of the studied sample ETZ, in lower proportion in ETZF-1 and ETZF-2 and seems not to be present in the crystallization process of the ETZF-5 glass sample. The lattice of this new allotropic phase γ-TeO_2 proposed recently by Champarnaud-Mesjard et al. can be considered as a chain system

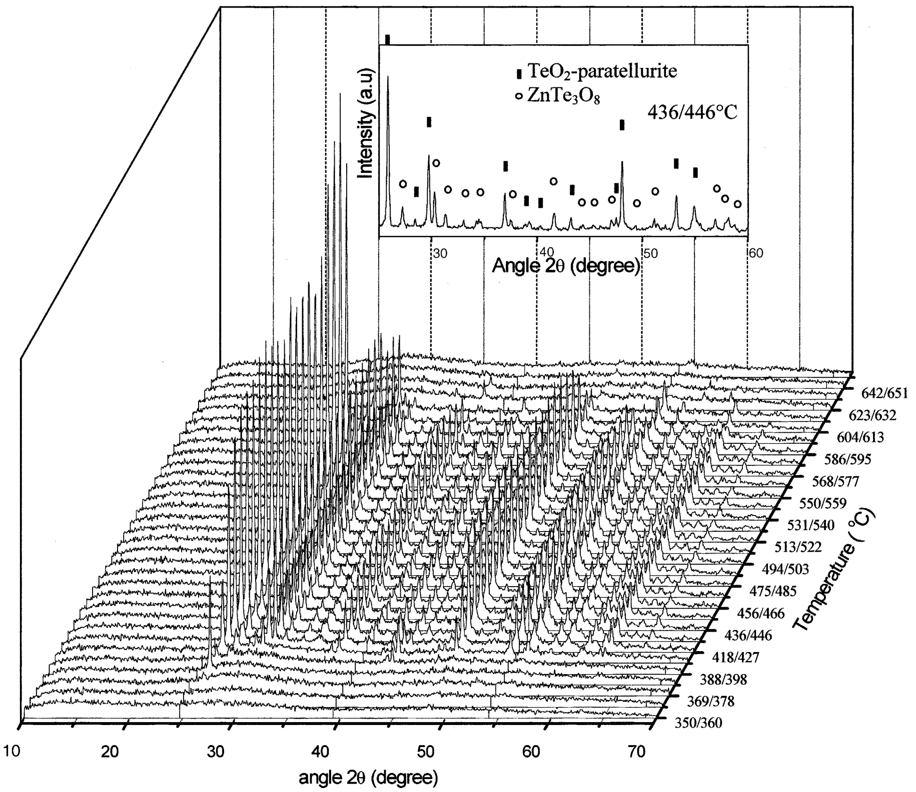


Fig. 3. X-ray diffraction patterns at various temperatures (from 350 to 650 °C, heating rate 10 °C/min) for the oxyfluoride zinc tellurite glass (ETZF-1) (■ α-TeO_2, ○ $ZnTe_3O_8$).

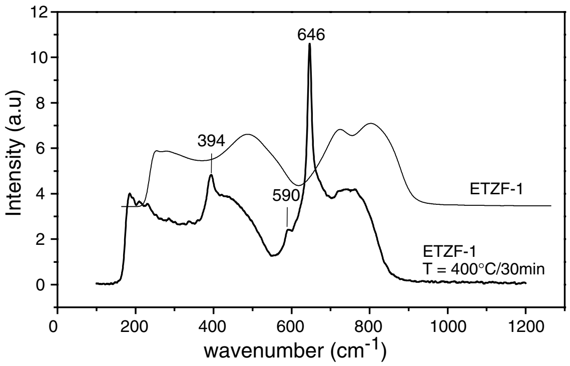


Fig. 4. Raman spectra of ETZF-1 glasses submitted to a heat treatment during 30 min at 400 °C and untreated.

[8]. In the case of $75TeO_2$–25ZnO, this phase appears since beginning of crystallization process about 383 °C and decomposed above 508 °C. For the two samples containing fluorine ETZF-1 and ETZF-2, the stability of this γ-TeO_2 phase is reduced and appears at slightly higher temperatures and decomposed at lower temperatures. The intensity of γ-TeO_2 specific peaks ($2\theta = 29.13°$, 27.2°, 27.08°) decreases whereas those of α-TeO_2 increase in intensity. The α-TeO_2 paratellurite crystalline phase appears at same temperature as γ-TeO_2, but the decomposition of this phase occurs at higher temperature. The three observed main peaks ($2\theta = 25.88°$, 29.72°, 48.08°) characteristic of α-TeO_2 gradually increase in intensity with temperature. They disappear above 600/630 °C for ETZ, ETZF-1 and ETZF-2. The phase $Zn_2Te_3O_8$ appears around 400 °C, gradually increases in intensity, and disappear above 600/630 °C for ETZ, ETZF-1 and ETZF-2. In the case of ETZF-5, the phase $ZnTeO_3$ clearly appears in agreement with the analysis of the phase diagram versus Zn/Te ratio. The $Zn_2Te_3O_8$ presence is also observed since the beginning of crystallization (373 °C). The crystallization of α-TeO_2 occurs at higher temperature than for the $75TeO_2$–$(25 - x)$ZnO–ZnF_2 glasses ($x = 0, 7, 16$). An unknown crystalline phase is also present appearing at the same temperature as $ZnTeO_3$ but disappears at higher temperature as α-TeO_2. This unknown phase (main peak at $2\theta = 31.1°$) seems to be also present in the crystallization of the glass $70TeO_2$–30ZnO [27]. The phase Zn_2TeO_3 and the unknown phase seems to dominate the first process of crystallization until 500 °C, while $Zn_2Te_3O_8$ peaks increase in intensity. At higher temperature, the main peaks are attributed to α-TeO_2, $ZnTe_3O_8$ and unknown phase. Above 593 °C, new three peaks appears without possible indexation.

3.3. Optical characterization

The UV–visible and near-infrared absorption spectra were measured for the studied series of glasses undoped and doped with Er^{3+}. The ground state transition energies of the Er^{3+} ions were determined in the glasses by absorption spectroscopy. The assignment of the absorption bands are illustrated in Fig. 5 that shows a typical spectrum from the sample ETZF-2. The absorption edge in the UV–visible spectral ranges due to electron transition transfer in host glasses is shifted from about 357 to 305 nm (Table 2). In the fluoride richest glasses, the slight blue shift of the band gap energy allows a better observation of the higher energy f–f transitions, which could be partially covered up in the band gap depending on the compositions of the glasses. The absorption cross-section of the oxide and oxyfluoride glasses, calculated from the experimentally measured absorption coefficient and the Er^{3+} concentration in the glass (Fig. 6), present a light decrease while the

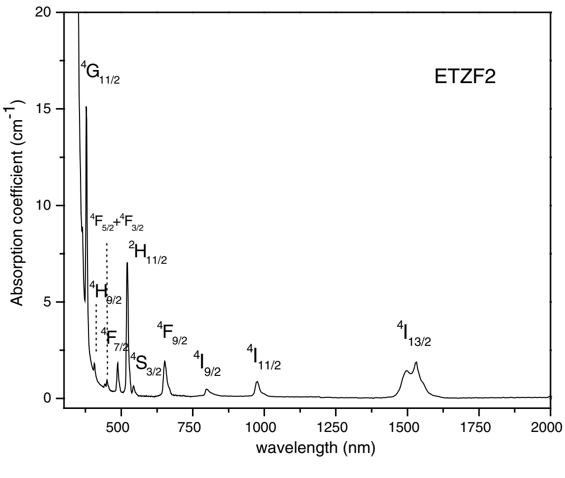


Fig. 5. Spectral absorption curves for the studied zinc tellurite oxyfluoride glasses of pUV–VIS–NIR.

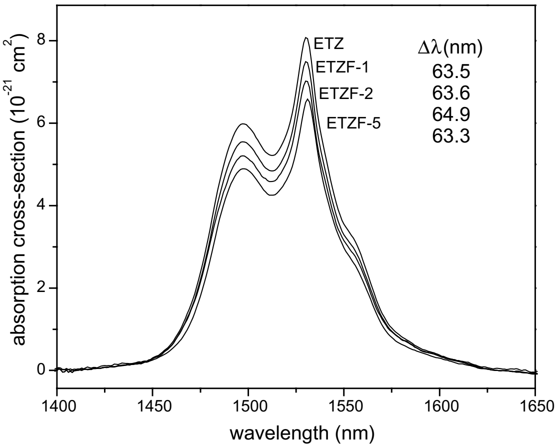


Fig. 6. Absorption cross-section around 1.5 μm band and effective absorption bandwidth (λ_{eff}) for oxide and oxyfluoride tellurite glasses.

fluorine substitution grows. The effective absorption bandwidth defined by $\Delta\lambda_{eff} = \int \sigma(\lambda)\,d\lambda/\sigma_{peak}$, where $\sigma(\lambda)$ is the absorption cross-section at the wavelength λ and $\sigma(\lambda_{peak})$ is the absorption cross-section value at the maximum peak, is about 63–65 nm for this serie of oxyfluoride tellurite glasses. The effective absorption bandwidth slightly increases with fluorine content until reach a maximum for ETZF-2 and decreases for ETZF-5 (Fig. 6). This feature follows the same trend as previously observed for lead halotellurite glasses [9].

From the absorption spectra transition, the experimental oscillator strengths f^{exp} are determined and can be express by this relation: $f^{exp}_{(J,J')} = mc/\pi e^2 N \int \alpha(v)\,dv$, where $\int \alpha(v)\,dv$ represents the integrated absorption coefficient $\alpha(v)$ (v: frequency (s^{-1})), J and J' are respectively the total angular momentum of the initial and the final state, N is the Er^{3+} ion concentration per volume unit, m is the electron masse, e the electron charge, h the Plank's constant and c the speed of light. The 10 first transitions considered from the ground level to the exited levels of Er^{3+} were $^4G_{11/2}$, $^2H_{9/2}$, $^4F_{3/2}$, $^4F_{5/2}$, $^4F_{7/2}$, $^2H_{11/2}$, $^4S_{3/2}$, $^4F_{9/2}$, $^4I_{9/2}$, $^4I_{11/2}$. Judd–Ofelt theory allows the calculation of the radiative and non-radiative probabilies and branching ratio of the transitions for rare earths in a variety of matrices [10]. The host-dependent parameters Ω_t are calculated by a least-square fit of calculated and experimental absorption oscillator strengths. Judd–Ofelt theory supposes an equipartition of the populations in the Stark sub-levels: the absorption spectra were measured at room temperature and under this condition, the Stark sub-levels of the ground state are thermally populated. The $4f$ transitions in the case of Er^{3+} are dominated by electric-dipole transition except for the $^4I_{15/2}/^4I_{13/2}$ transition which has a magnetic-dipole (MD) component. The oscillator strength of this MD transition was obtained using the procedure of the Ref. [11]. So, magnetic dipole contributions were subtracted from the experimental oscillator strengths in the case of the $^4I_{15/2}/^4I_{13/2}$ transition to obtain the electric-dipole contribution. The Judd–Ofelt (J–O) parameters were then calculated by a least square fitting of calculated and experimental oscillator strengths. The calculated oscillator strengths $f^{calc}(J' \to J)$ can be express by this expression $f^{cal}_{(J,J')} = (8\pi^2 m v)/3h(2J+1)e^2 n^2(\chi_{ed}S_{ed}(J,J') + \chi_{md}S_{md}(J,J'))$ for a transition of average frequency v, where n is the linear refractive indices considered as a constant in the calculation and measured at 1064 nm, where the χ terms account for a local field correction for Er^{3+} ions, S_{md} is a magnetic dipole line strength. The electric dipole line strength S_{ed} is given by $S^{calc}_{ed}(J \to J') = \sum_{t=2,4,6} \Omega_t |\langle J\|U^{(t)}\|J'\rangle|^2$. The double reduced matrix elements $\langle\|U^{(t)}\|\rangle$ of unit tensor $U^{(t)}$ of rank t calculated by Carnall were assumed to be constant from host to host [12]. The least square fit in the Judd–Ofelt calculations results in minimization of the root mean square (RMS) deviation

$$\mathrm{RMS} = \sqrt{\frac{\sum_i (f_i^{exp} - f_i^{calc})^2}{M-3}},$$

where M is the number of transitions used for the fitting ($M = 10$) subtracted by the number of fitting parameters. The estimation of the radiative rates A_{ij}, the branching ratio β_{ij} and the radiative lifetime were obtained using the Judd–Ofelt parameters and the following formula:

$$A_{ed}(J' \to J) = \frac{64\pi^2 e^2}{3h(2J'+1)} \frac{n(n^2+2)^2}{9\bar{\lambda}^3} S^{calc}_{ed}(J' \to J)$$

and for the md transition,

$$A_{\mathrm{md}}(J' \to J) = \frac{64\pi^2 e^2}{3h(2J'+1)} \frac{n^3}{\bar{\lambda}^3} S_{\mathrm{md}}^{\mathrm{calc}}(J' \to J).$$

The radiative lifetime τ_{rad} and the branching ratio β are then obtained by these relations:

$$\tau_{\mathrm{rad},J'}(J' \to J) = \frac{1}{A_{J'}} = \frac{1}{\sum_{J'} [A_{\mathrm{ed}}(J' \to J) + A_{\mathrm{md}}(J' \to J)]}$$

$$\text{and} \quad \beta(J,J') = \frac{A_{\mathrm{rad}}(J,J')}{W_{\mathrm{R}}}.$$

As an example, the comparison between the experimental and calculated oscillator strengths in the case of the $75TeO_2$–$25ZnO$ and $75TeO_2$–$17ZnO$–$8ZnF_2$ is reported in Table 4. The calculated Judd–Ofelt parameters of the different glasses are reported in Table 5 with the root mean square deviations, refractive indices and the concentration of erbium ions per volume unit. The RMS deviations are comparable to deviation obtained with Er^{3+} ions in tellurite glasses [13,14]. The radiative decay rates, branching ratios and radiative lifetimes of several Er^{3+} emission transition are given in Table 5. The values of the *J–O* parameters appear to be in the range usually observed for oxide, oxyfluoride tellurite and oxyfluoride glasses [9,15–17].

The tellurite glasses have a broad IR transmission spreading up to 6 μm, but the presence of hydroxyl groups impairs the optical characteristics of tellurite glasses in the middle IR range. On the spectra represented in Fig. 7, the absorption peak were normalized with respect to the thickness of the glass samples. A broad absorption band 2500–3600 cm^{-1} and a weaker at 2260 cm^{-1} (4.4 μm), can be distinguished and assigned to the stretching vibration of OH^- groups.

The normalized measured emission spectrum of oxide and oxyfluoride tellurite glass are shown in Fig. 8 illustrating the broad emission spectra around 1.5 μm in these tellurite glasses. The global shape of the peak emission is modified by the different contributions of each components included into the broad emission band. In our oxyfluoride samples, the shoulder at lower wavelengths beside the maximum of the peak around 1530 nm slightly decreases. The main shoulder

Table 4
Experimental and calculated oscillator strengths of Er^{3+} absorption transitions from the ground state $^4I_{15/2}$ for ETZF-1 and ETZ (A_w: average wavelength, A_f: average frequency, D: deviation)

Excited state	A_w (nm)	A_f (cm^{-1})	f_{exp} (10^{-8})	f_{calc} (10^{-8})	D (10^{-8})
$75TeO_2$–$25ZnO$					
$^4I_{13/2}$	1520	6579	260	165 (ed) 63 (md)	32
$^4I_{11/2}$	979	10 215	85	80	5
$^4I_{9/2}$	803	12 453	55	53	6
$^4F_{9/2}$	655	15 267	323	320	3
$^4S_{3/2}$	544	18 382	60	58	2
$^2H_{11/2}$	521	19 194	1301	1300	1
$^4F_{7/2}$	489	20 450	246	262	−16
$^4F_{3/2}$ et $^4F_{5/2}$	448	22 321	106	111	−5
$^2H_{9/2}$	407	24 570	71	91	−20
$^4G_{11/2}$	377	26 525	2290	2291	−1
$75TeO_2$–$17ZnO$–$8ZnF_2$					
$^4I_{13/2}$	1519	6583	179	153 (ed) 62 (md)	26
$^4I_{11/2}$	978	10 225	73	73	0
$^4I_{9/2}$	803	12 453	50	48	2
$^4F_{9/2}$	653	15 314	299	295	4
$^4S_{3/2}$	544	18 382	59	56	3
$^2H_{11/2}$	521	19 194	1058	1089	−31
$^4F_{7/2}$	489	20 449	229	246	−17
$^4F_{3/2}$ et $^4F_{5/2}$	448	22 321	101	106	−5
$^2H_{9/2}$	407	24 570	85	86	−1
$^4G_{11/2}$	377	26 525	1933	1915	18

Table 5
Judd–Ofelt parameters Ω_2, Ω_4, Ω_6, RMS deviation, refractive indices and the concentration of Er (ions/cm^3) of the oxide and oxyfluoride tellurite glasses

Glass matrix	ETZ	ETZF-1	ETZF-2	ETZF-5
N ($\pm$0.02)	2.03	2.00	1.96	1.80
ρ ($\pm$0.01 g/cm^3)	5.55	5.51	5.48	5.38
Er (10^{20} ions/cm^3)	2.6	2.7	2.7	2.7
Ω_2 (10^{-20} cm^2) $\pm$ 10%	5.60	4.71	3.92	2.82
Ω_4 (10^{-20} cm^2) $\pm$ 10%	1.87	1.74	1.58	1.38
Ω_6 (10^{-20} cm^2) $\pm$ 10%	1.03	1.01	0.94	0.95
$\sum \Omega_t$ (10^{20} cm^2)	8.5	7.5	6.4	5.2
RMS (10^{-8}) $\pm$ 5%	15.8	16.1	15.4	20.2

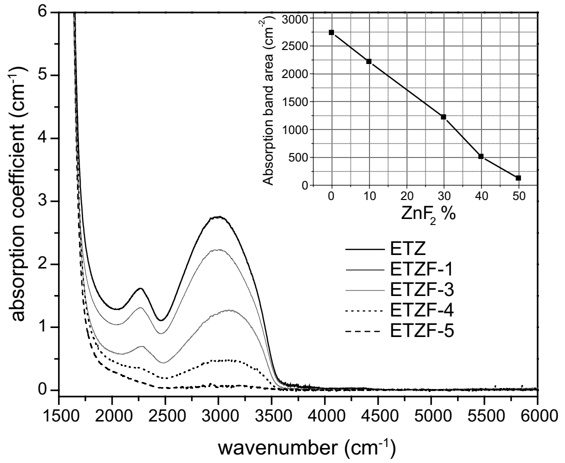


Fig. 7. Infrared spectra for the oxide and oxyfluoride tellurite glasses.

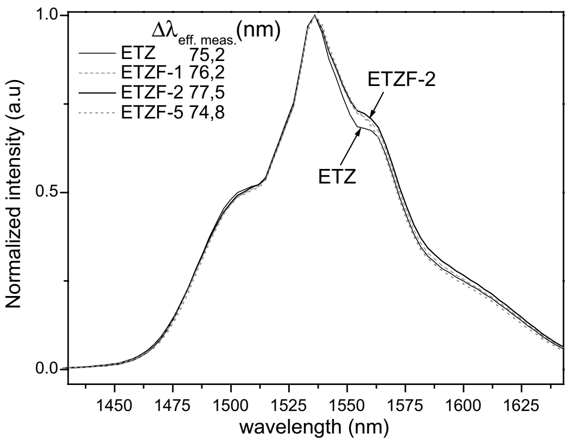


Fig. 8. Normalized emission intensity around 1.5 μm band for oxide and oxyfluoride tellurite glasses and effective emission linewidths ($\Delta\lambda_{\mathrm{eff.meas.}}$).

around 1560 nm is more intense in relative intensity than those observed on oxide glasses with similar erbium contents. This feature could support a slight flattening of the maximum gain in the range of 1535–1565 nm compared to tellurite glasses. In the case of the fluorine richest glass, the whole profile of emission peak seems to remain unchanged, similar to the other oxyfluoride samples, with the exception of a light thinning of the emission band supplementary observed. The effective emission linewidths for the normalized measured emission spectra have been calculated and the ETZF-1 and ETZF-2 oxyfluoride tellurite glasses show a wider emission linewidths than oxide tellurite glasses. The comparison of 2.7×10^{20} and 3×10^{19} ions/cm^3 erbium concentration clearly shows a modification of the emission spectrum shape and the broadening observed for the richest dopant glasses relies on the higher intensity of the main shoulder at 1560 nm (Fig. 9).

The lifetime measurements of the $^4I_{13/2}$ excited level were achieved for two concentrations of Er^{3+} in order to compare the possible effect of self-quenching due to more probable interaction between two neighboring erbium ions. The emission lifetimes were determined by a least squares fit a single exponential and no difference between measurements was noticed between bulk and powder samples (Fig. 10). The experimental lifetimes values obtained for tellurite glasses ($75TeO_2$–$20ZnO$) are around 3.2 ms in the same order than tellurite systems reported [5,9]. When the proportion of fluorine increases, the lifetime of both $^4I_{13/2}$ and $^4I_{11/2}$, significantly increases. The effects of substitution of GeO_2 to TeO_2 on lifetime decay are studied. Within our measurement accuracy, a

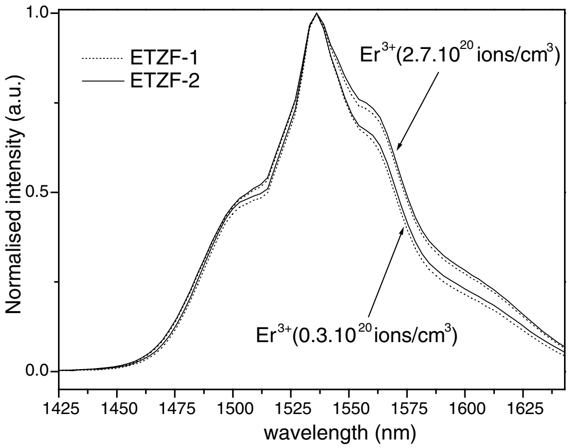


Fig. 9. Normalized emission intensity around 1.5 μm band for oxyfluoride tellurite glasses ETZF-1 and ETZF-2 for two concentration of erbium (0.3×10^{20} and 2.7×10^{20} ions/cm^3).

positive effect on the $^4I_{13/2}$ level lifetime takes place if the GeO_2 concentration remains lower than 10% molar, while the $^4I_{11/2}$ level lifetime decreases slightly. By recording the emission at 1000 and 1530 nm under pumping at 800 nm corresponding to the radiative transition $^4I_{11/2} \rightarrow {}^4I_{15/2}$ and $^4I_{13/2} \rightarrow {}^4I_{15/2}$ respectively, the effect of the GeO_2 introduction is also apparent on the ratio of this two spontaneous luminescence intensity. Compared to the former composition $75TeO_2$–$9ZnO$–$16ZnF_2$, this ratio $R = I_{11/2}I_{13/2}$ is divided by a factor 1.5 for a substitution of 5 mol% GeO_2 and 1.65 for a substitution of 10 mol% GeO_2.

The evolution of emission spectra in the range 400–750 nm for these sample recorded at room temperature and pumped at 974 nm are shown in Fig. 11. A possible up-conversion mechanism of Er^{3+} by 974 nm is shown in Fig. 12. The dominant emission is in the green corresponding to $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transition of Er^{3+} at 548 nm. A red emission band at 660 nm and a weaker at 525 nm

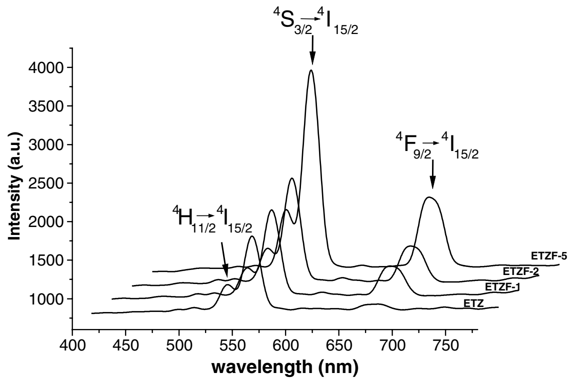


Fig. 11. Emission spectra in visible range of oxide and oxyfluoride tellurite glasses.

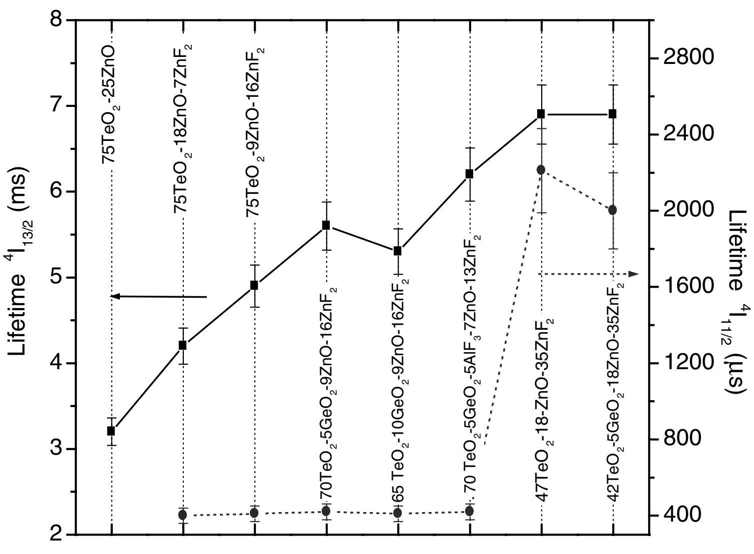


Fig. 10. Evolution of the $^4I_{11/2}$ and $^4I_{13/2}$ lifetime through several composition of oxide and oxyfluoride tellurite glasses.

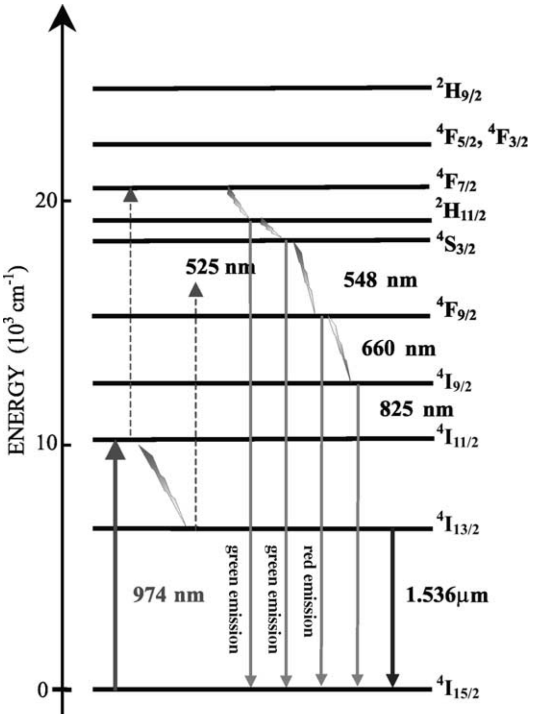


Fig. 12. Schematic representation in simplified energy diagram of radiative (black arrow for 1.53 µm and grey) and non-radiative (gradual grey arrows) processes occurring in tellurite glasses pumped at 980 nm.

are also observed on the spectra associated to $^4F_{9/2} \rightarrow {}^4I_{15/2}$ and $^4H_{11/2} \rightarrow {}^4I_{15/2}$ respectively. According to the Fig. 11, the up-conversion efficiency increases with increasing fluorine content and decreasing TeO_2. For oxyfluoride tellurite glasses, the $^4F_{9/2} \rightarrow {}^4I_{15/2}$ emission is more notable in relative proportion to green luminescence compared to tellurite oxide glasses.

4. Discussion

4.1. Glass structure

As fluoride coumpounds is melted with TeO_2, evaporation of fluorine during the melting process is more or less unavoidable under the main form of HF following this oversimplified reaction $OH^- + F^- \rightarrow HF + O^{2-}$. Even with a rapid melting using a crucible covered with a plate to prevent the fluorine volatilization tendency, a systematic loss of about 1–4 wt% has been observed for the fluorine confirmed by different analysis technique. In the case of ETZF-5, the light tellurium loss can be explained by a probable evaporation in observable proportion of high volatility compounds like TeF_4. The decrease of the glass transition temperature evolution observed on DTA curves is due to a gradual introduction of ($ZnO + ZnF_2$) in higher proportion and consequently, the fluorine atoms introduced in oxide network can be considered not to act as a bridge between two Te atoms. Breaking the oxide network, the crosslinking degree of the glass former could consequently be reduced. This behavior induced by the addition of fluorine is strengthened by an additive effect due to a Zn/Te ratio growing. Furthermore, it was also observed that the introduction of erbium in growing proportion results in a slight increase about 5/10 °C of the glass transition temperature of our oxide and oxyfluoride glasses. This behavior already reported can be explained by an increase of the rigidity of the network strengthened by the presence of erbium which can partially participate to the formation of the glass network [18].

4.1.1. Raman analysis

The structure of tellurite glasses is mostly made of asymmetrical unit structure of trigonal bipyramids TeO_4(tbp) with a lone pair of electron in an equatorial position and linked each other by a common oxygen in axial position in one polyhedron and equatorial position for the second polyhedron forming Te_{ax}–O_{eq}–Te linkages as in three-dimensional network of α-TeO_2 crystal [19]. The introduction of a modifier in growing proportion leads to the presence of TeO_3 trigonal pyramids (tp) through an intermediate stage formation of TeO_{3+1} polyhedrons. These coordination changes of the tellurium atom involves a progressive elongation of a Te–O bond in axial position from the initial trigonal bipyramids TeO_4 and an increase of non-bridging oxygen atoms [20]. The proportion of TeO_3 and TeO_{3+1} entities within the former network depends on the chemical nature and concentration of modifier [21]. The bands around 775 and 735–750 cm^{-1} are assigned, in previous studies [22,23], to stretching vibrations mode mainly of TeO_{3+1} and TeO_3 entities involving non-bridging oxygen (NBO) and also TeO_4 entities. The band around 660 cm^{-1} is assigned to antisymmetric vibrations of Te–O–Te linkages combining TeO_4, TeO_{3+1} and TeO_3. According to published studies, this band involving two dissymmetric bonds Te-O, a long and a short length, characterizes the degree of connectivity of the glass network [22]. The increase of the $[ZnF_2 + ZnO]/[TeO_2]$ ratio results in a significant modification of the Raman spectra shape and the intensity ratio $(I_{740}+I_{775}/I_{660})$ of these bands continuously evolves. This observation mainly indicates a decrease of the TeO_4 entities involving the formation of more TeO_{3+1} and/or TeO_3 groups. The obvious intensity increase of the band around 775 cm^{-1} confirms the formation of TeO_{3+1} and TeO_3 with non-bridging oxygen. Since the Raman vibrations are mainly sensitive to the polarizability of the tellurium entities, the change induced by a likely formation of oxyfluoride tellurite entities should be affected the Raman spectrum feature. Some O^{2-} atoms might be effectively replaced by F^- because of their comparable ionic radius and formally, one oxygen may be replaced by two fluorine breaking the former network. At constant ratio $[ZnF_2 + ZnO]/[TeO_2]$, these probable changes remain soft and can be due to a low proportion of oxyfluoride tellurite entities which their distinctive Raman signatures could be hidden within the broad bands in large range of wavenumbers (600–800 cm^{-1}). However, the shoulder observed at higher wavenumber (around 775 cm^{-1}) for growing fluorine concentration, in agreement with Sekiya et al., is attributed to Te–O^- non-bridging stretching vibration mode of TeO_{3+1} and TeO_3 groups [22]. In parallel, the intensity of the band located at 420 cm^{-1}, related to the bending vibration modes of Te–O–Te linkages, slightly decreases. This effect illustrates the decrease of the Te_{eq}–$O\cdots Te_{ax}$ linkages of a continue network of TeO_n ($n = 4, 3+1$ and/or 3) entities which is consistent with the previous observations.

4.1.2. Crystallization of oxyfluoride glasses

The X-ray diffraction data were analyzed based on the knowledge of the phase diagram of the system ZnO–TeO_2 which presents two crystalline phases $Zn_2Te_3O_8$ [24] constituted of elementary unit ($Te_3O_8^{4-}$) formed by connecting one TeO_4(tbp) (with two NBO atoms in equatorial position) with two TeO_{3+1} entities (with one NBO atoms) and $ZnTeO_3$ composed of (TeO_3^{2-}) isolated [25]. Usually the description of the tellurite glass structure is related to the crystalline polymorphs α- and β-TeO_2 consists of TeO_4(tbp) [19,26]. A structural organization of tellurite glasses in this system has been proposed from infrared and Raman spectroscopy studies, NMR, neutron or X-ray diffraction studies [6,22,23,27]. A phase diagram study of TeO_2–TeF_4 system has also been proposed with a large glass-forming range and two crystalline oxyfluoride phases $TeOF_2$ and $Te_2O_3F_2$. The structure of the crystal $TeOF_2$ and TeO_3F_2 were related to the α- and β-TeO_2 polyforms, respectively [28]. The equatorial positions are occupied by bridging oxygen atoms whereas fluorine atoms in axial positions are comprehensively non-bridging. The cohesion of the network is partially guaranteed by weak Te–F bonds between more covalent sheets ($Te_2O_3F_2$) or chains ($TeOF_2$).

We analyzed the crystallization in relationship with DTA analysis. For studied glass samples, the crystallization process begins at 383 °C (ETZ, ETZF-1), 398 °C (ETZF-2), 373 °C (ETZF-5) within our experimental accuracy (Table 3). These values are closed, but slightly inferior for ETZ and ETZF-1, to the peak crystallization observed by DTA analysis in which onset values T_x are around 368–404 °C depends on composition. For the same Te/Zn ratio (ETZ, ETZF-1 and 2), the temperature of starting crystallization does not follow the same trend as crystallization peak temperature with raising fluorine content. Indeed, the crystallization process seems to be less easier with introduction of fluorine in accordance with the slight increase of the difference $T_x - T_g$ observed by DTA. On the contrary, the crystallization process in ETZF-5 is going along the decrease of the crystallization peak temperature recorded by DTA constantly showing a different behavior. The DTA analysis of ETZF-2 and ETZF-5 sample present a main crystallization peak with a shoulder at higher temperature. This feature results in a spreading of the crystallization temperature range illustrated by a broad crystallization peak particularly noticed for ETZF-2 in which a decline in magnitude is additionally observed. This crystallization peak feature of ETZF-2 might be related to the beginning of crystallization occurring at higher temperature. According to the DTA and X-ray diffraction experiments, the large exothermic crystallization peak observed for the three first samples seems to correspond to the formation of α-TeO_2 and γ-TeO_2 including $Zn_2Te_3O_8$; whereas $ZnTeO_3$ seems to be related to a lower temperature crystallization peak. An accurate XRD experiment should be doing around the temperature range of 350/450 °C in order to determine more precisely the temperature of the crystallization beginning of those phases.

The main building units of the initial crystal structures which appear at lower temperature of crystallization persist to a great extent in the vitreous state as at short distance ordering of atoms in glasses. For ETZ, ETZF-1 and ETZF-2, the first-crystallization of α-TeO_2, γ-TeO_2 and $Zn_2Te_3O_8$ phases is the probable signature of the presence in the former network of both coordination TeO_4 and TeO_{3+1}. According to Raman results, these glasses can be described as a partial continuous network structure composed mainly of TeO_4(tbp) and TeO_{3+1} polyhedra like in γ-TeO_2. The structural organization of the glass should also include a distribution of $(Te_3O_8^{4-})_n$ finite chains and the lower coordination tellurium group TeO_{3+1} with one non-bridging oxygen might be related to the surrounding presence of Zn^{2+} atoms allowing an easier crystallization of $Zn_2Te_3O_8$. The ETZF-5 glass structure network according to X-ray diffraction studies and Raman spectroscopy seems to be strongly different. The crystallization of $ZnTeO_3$ should indicate the presence of isolated structural fragments, such as TeO_3^{2-} are formed within the glass network. However, the presence of Zn_2TeO_8 at the beginning of the crystallization process and the appearance of α-TeO_2 crystalline phase in smaller proportion at higher temperature range agree with the results obtained by Raman which predict a partial preservation of higher coordination for the tellurium tbp(TeO_4). The length of the $(Te_3O_8^{4-})_n$ chains with TeO_3^- as terminal groups should decreases as the ratio of $(ZnO_2 + ZnF_2)/TeO_2$ increases and the fluorine substitution resulting in the presence of some TeO_3^{2-} isolated groups likely present in ETZF-5 glass.

The lack of X-ray diffraction peaks signature of fluoride or oxyfluoride crystalline phases, like TeF_4, $TeOF_2$, $Te_2O_3F_2$ for mentioning known crystalline phases in this system, could mean that oxyfluoride tellurite entities are not present in strong proportion. However, these phases decompose at 129, 169, 352 °C respectively and in our experimental conditions, an observation of their crystallization seems not to be probable. The ZnF_2 compound which is a better glass former than ZnO can favor the glass arrangement in larger range than tellurite glass by attending to the glass network formation. Nevertheless, the crystallization of ZnF_2 is not clearly observed in our experimental setting. Thus, the reaction between ZnF_2 and TeO_2 during the melting remains strongly probable as the fluorine concentration increases and leads to the formation of tellurium and zinc oxyfluoride. If we consider that some O^{2-} of the Te polyhedra is likely replaced by F^- non-bridging, the coordination polyhedra of Te^{IV} should be closed to those encounter in oxyfluoride tellurite crystalline compounds. Te^{IV} is quasi-systematically fivefold coordinated in pure fluoride compounds as for example TeF_4 [29], fivefold/fourfold in $TeOF_2$ and $Te_2O_3F_2$. In the case of oxide, Te is fourfold coordinated (α and γ-TeO_2), fourfold and (3+1) in $ZnTe_3O_8$ and treefold coordinated in $ZnTeO_3$. Therefore, various coordination can be expected in oxyfluoride glasses. But, the influence of fluorine to increase the coordination number of Te^{IV} could be overlooked by the coordination decreasing accompanying the ratio ZnF_2/TeO_2 raising cleaving the Te–O–Te linkages. Moreover, the introduction of fluorine has a tendency to prevent the γ-TeO_2 allotropic phase formation and to shift the crystallization of $Zn_2Te_3O_8$ and α-TeO_2 to little higher temperature. These observations are consistent with the idea that the introduction of zinc fluorine results in a depolymerization of the tellurite network which affects the stability of γ-TeO_2 and the crystallization ability of α-TeO_2 and $Zn_2Te_3O_8$ by cleaving the Te_{ax}–O_{eq}–Te linkages and partial forming of tellurium and zinc entities with mixed anion, fluorine and oxygen.

4.2. Optical characterization

4.2.1. Judd–Ofelt analysis

The physical meaning of the Judd–Ofelt parameters has been discussed by several authors. The Ω_2 parameter is sensitive to the symmetry of the rare earth site and the covalency between rare-earth ions and ligand anions whereas the physical meaning of Ω_4 and Ω_6 is more unclear [30]. The rigidity of the network surrounding the rare-earth ion was suggested to influence the magnitude of Ω_4 and Ω_6 [31]. Tanabe et al. showed that, for oxide and fluoride glasses, the effect of host glass on Ω_6 can been explained in term of local basicity change of the ligands [32]. We can mention a continuous diminution of the Ω_2 Judd–Ofelt parameters with the increasing amount of fluorine within the host matrix. The value of Ω_2 can be expected to be more important in oxide glasses than in fluoride glasses [17]. The decrease of Ω_2 involving by the fluorine content follows this trend and can be explained by the predominant decrease of the covalency degree in rare earth site in accordance with Ebendorff et al. in oxyfluoride phosphate glasses [33]. The growing presence of fluorine within the host matrix, which presents a relative small polarizability, decreases the global optical basicity of the network. As the ZnF_2 content increases, the presence of fluorine ions around Er^{3+} should be more probable resulting in a greater decrease of the covalency between Er^{3+} and ligand anions. The increase of the ratio Te/Zn for ETZF-5 could also affect the Ω_t parameters; indeed, the increase of ZnO and ZnF_2 proportion leads to the alteration of the tellurium network in which the ionic character increases. Moreover, the entities TeO_4 are continuously changed in TeO_{3+1} and TeO_3 with an expected lower polarizability than TeO_4 groups [34]. This global evolution through studied glasses confirmed by the blue shift of the cut-off wavelength which roughly depends on the electronegativity difference between anionic and cationic constitutive elements of the former network and on the polarization of anionic electronic density by the cation. In the case of our glasses series, the change observed for Ω_6 do not show a significant change only a slight decreasing while fluorine content grows. The global decrease of the sum Judd–Ofelt parameters with increasing proportion of fluorine is in accordance with reported results which predicted that the sum of the Judd–Ofelt parameters increases in the order fluoride < oxyfluoride < oxide glasses mostly due to the decrease of covalency of the chemical bond between the rare-earth ion and the ligand anions [17]. In conclusion, the Judd–Ofelt parameters sum values of studied glasses lie between those of fluoride and oxide glasses depending on the F/O ratio in our glasses.

The calculated radiative decay rates of Er^{3+} emission transition depend on the value of Ω_t parameters and among them mostly by Ω_6. Moreover, the observed refractive indices gradually reduces with at once introduction of fluorine and ratio ZnO/TeO_2 growth as it was expected for lessening the entity proportion presenting a high polarizability. The continuous refractive indices lowering coupled with the J–O parameters decline leads to an increase of the radiative decay rates as observed experimentally for Nd^{3+} fluorotellurite glasses [4]. The results are reported in Table 6.

4.2.2. IR spectroscopy

The damageable quenching effect of OH upon the $^4I_{13/2}$ levels of Er^{3+} have been reported in several systems like phosphate or tellurite glasses [5,35,36]. As the fundamental stretching vibration mode of OH group is included between 2500 and 3600 cm^{-1}, the process of non-radiative relaxation of the excited level ($^4I_{13/2}$, $^4I_{15/2}$) can involve only two phonons. In Fig. 10, the absorption band closed to 3.2–3.5 μm is due to the stretching vibration of free OH^- groups and the absorption in the lower range of energy is supposed to correspond to the presence of stronger hydrogen bond –O–H···O– as observed in tellurite or phosphate glasses [37,38]. The spreading of this absorption band related to OH vibration illustrates an important distribution of hydroxyl oscillator in the tellurite matrix where the hydroxyls groups surrounding is greatly diversified. The introduction of fluoride compounds into batch is an efficient method to remove OH^- hydroxyl in tellurite glasses. The hydroxyl and fluorine ions are isoelectronic with a similar ionic size, so that hydroxyl ions can be easily replaced by fluorine

Table 6
Relative decay rate A_{ij}, branching ratio β_{ij}, calculated radiative lifetime τ_{rad} of the $^4I_{13/2}$ and $^4I_{11/2}$ excited level of Er^{3+}

Glass matrix	Initial state	Final state	A_{ed} (s^{-1})	A_{md} (s^{-1})	β_{ij}	τ_{rad} (ms) ± 10%
ETZ	$^4I_{13/2}$	$^4I_{15/2}$	224.2	85.1	100.0	3.2
	$^4I_{11/2}$	$^4I_{15/2}$	304.4		84.3	2.8
	$^4I_{11/2}$	$^4I_{13/2}$	39.9	16.7	15.7	
ETZF-1	$^4I_{13/2}$	$^4I_{15/2}$	204.3	81.4	100.0	3.5
	$^4I_{11/2}$	$^4I_{15/2}$	270.7		83.9	3.1
	$^4I_{11/2}$	$^4I_{13/2}$	35.9	16.0	16.1	
ETZF-2	$^4I_{13/2}$	$^4I_{15/2}$	175.5	76.6	100.0	4.0
	$^4I_{11/2}$	$^4I_{15/2}$	228.4		83.3	3.6
	$^4I_{11/2}$	$^4I_{13/2}$	30.6	15.1	16.7	
ETZF-5	$^4I_{13/2}$	$^4I_{15/2}$	128.0	59.3		5.3
	$^4I_{11/2}$	$^4I_{15/2}$	161.2		82.6	5.1
	$^4I_{11/2}$	$^4I_{13/2}$	22.2	11.9	17.4	

during melting. Therefore, the decreasing of both absorption bands on IR spectra can be interpreted on the basis of a concentration increase of introduced fluorine.

4.2.3. Luminescence properties

4.2.3.1. Composition dependence on lifetime. In the limit of a weak excitation, the lifetime for an excited level of a rare-earth ion is governed by the following expression: $1/\tau = 1/\tau_R + 1/\tau_{NR}$, where τ_R, and τ_{NR} is the radiative and non-radiative decay lifetime respectively. The radiative decay lifetime is usually estimated from the Judd–Ofeld method calculation. The estimation error through Judd–Ofelt analysis was reported no larger than about 20% [39]. In the case of tellurite glasses, the observed decay lifetime is in the same order of magnitude of the value calculated by Judd–Ofelt method, suggesting that the influence of multiphonon relaxation seems to be minor. For oxyfluoride tellurite glasses, we observed a difference between the radiative lifetime determined by calculation and observed experimentally. An observed lifetime longer than a calculated one is not unexpected on oxyfluoride glasses and this behavior, which is not entirely understood yet, has already been reported [36]. Additionally with error caused by calculation, some self-absorption effects might be occurring for the oxyfluoride samples. An absorption of a photon emitted from an Er^{3+} ion contributes to populate the emitting level of the non-excited Er^{3+} ion and, consequently, increases the apparent experimental lifetime. In order to avoid or reduce the re-absorption phenomena, the beam was focused to the surface of the bulk sample. Thus, the evolution of lifetime induced by the introduction of fluorine can be analyzed qualitatively. The trend expected from the estimation of the decay lifetime by Judd–Ofelt method calculation is experimentally confirmed and illustrated by the Fig. 4. Indeed, the experimental lifetime is regularly increased with a fluorine content growth. The non-radiative decay lifetime $\tau_{NR}^{-1} = W_{MP} + W_{ET}$ – with multiphonon decay rate (W_{MP}), non-radiative energy transfer rate between active ions (W_{ET}) – is due predominantly to multiphonon relaxation in absence of significant concentration quenching effects. Relaxation of rare-earth ions through multiphonon processes is dominated by the energy gap ΔE between the two electronic levels and also depends on highest phonon energy of the host glasses and electron–phonon coupling strength [40]. Consequently, the low phonon energy typical of tellurite glasses and of our oxyfluoride glasses, can favor a quite long emission lifetimes of the $^4I_{13/2}$ level mostly governed by radiative decay rate. According to a lower impact of multiphonon loss, the lifetime values recorded for oxyfluoride glasses should be strongly influenced by the presence of fluorine within glass host and additionally, to the increase of the ratio Zn/Te in the ETZF-5 glass case. They strongly modify the local electrostatic field strength and symmetry of the ligand field around the rare-earth related as it was expected from structural organization analysis and predicted by J–O calculation trends. In other hand, the first overtone of the fundamental stretching vibration of OH groups (2500–3600 cm^{-1}) well matches to the energy gap of the $^4I_{13/2} \rightarrow {}^4I_{15/2}$ (around 6600 cm^{-1}). Thus, this particular multiphonon loss mechanism should be taking into account in the declining of radiative processes. Therefore, the $^4I_{13/2}$ lifetime is likely improved due to the decrease of OH group concentration observed by FT-IR spectroscopy with the growing fluorine substitution within the glasses.

As a consequence of the energy gap between $^4I_{11/2}$ and $^4I_{13/2}$ levels, the lifetime of the $^4I_{11/2}$ level is strongly influenced by the phonon energy. Therefore, the low phonon energy typical of tellurite glasses improves the lifetime of the $^4I_{11/2}$ level compared to silicate glasses, for example. Due to lower probability of non-radiative multiphonon relaxation to $^4I_{13/2}$ level, this behavior more marked as the fluorine content increases could reduce the efficiency of an amplifier. Indeed, both a pump excited state absorption mechanism (ESA) from the $^4I_{11/2} \rightarrow {}^4F_{7/2}$ and cross relaxation process can occur under 980 nm high pumping power. The choice of a direct pumping of the $^4I_{13/2}$ level by a 1480 nm pump can resolve this problem. In counter part, this pumping choice usually causes a degradation of the ratio signal-to-noise compared to the first alternative at 980 nm. Another more suitable alternative is to depopulate the $^4I_{11/2}$ level of Er^{3+} inducing the reduction of the up-conversion from this level. The $^4I_{11/2}$ level lifetime can be affected by introducing slight proportion of elements, like B_2O_3 and P_2O_5, having phonon overtones that are resonant with the energy difference between the two levels [41]. The presence of GeO_2 could play a similar role giving an highest phonon energy around 900 cm^{-1} for germanate glasses greater than tellurite glasses [42]. The effect of the GeO_2 introduction remain minor: the required number of phonons P for the relaxation from $^4I_{11/2}$ to $^4I_{13/2}$, determined by $P = \Delta E/\hbar\omega$, where ΔE is the energy difference between the two levels and $\hbar\omega$ phonon energy of host matrix, is still too elevated for influencing sharply the $^4I_{11/2}$ lifetime.

4.2.3.2. IR and visible emission. When pumped at 980 nm (Er^{3+}, $^4I_{15/2} \rightarrow {}^4I_{11/2}$ transition), the system should be compared to a three level scheme where the emission at 1530 nm corresponds to the transition $^4I_{13/2} \rightarrow {}^4I_{15/2}$. These two energy levels $^4I_{13/2}$ and $^4I_{15/2}$ are consisting of several Stark sub-levels and the emission peak and profile are influenced by the Stark splitting of both levels and oscillator strengths of the individual transitions between the different Stark sub-levels. Both are affected by the ligand field of the Er^{3+} environment but its influence on the shape of the emission peak at 1.5 μm is not clearly understood yet. Due to a crystal field change, the stark splitting is expected to decrease from oxide to fluoride glasses and a such behavior should have a continuous impact on the feature of the emission peak while increasing fluorine contents. Moreover for each rare-earth ion, the shape and magnitude of the emission peak is strongly influenced by the crystal field disparity from site-to-site (Fig. 8). Consequently for an analogous Er^{3+} concentration, an apparent inhomogeneous broadening, observed on Fig. 9, is mainly originated from the glass network structure variation from site-to-site. In the case of rich erbium concentration, the spectral broadening may be explained by differential site occupancy and also by a possible fluorescence emission reabsorption effect. The whole involved phenomena result in a complex feature of the emission peak depending on the contributions of each effects. Thus, the difference of the linewidth and emission peak shape between oxide and oxyfluoride tellurite glasses at 1.5 μm is due mostly by the difference in the local environment of the Er^{3+} ions. In tellurite glasses, the Er^{3+} ions have a coordination sphere of oxygen, whereas in oxyfluoride glasses they might be surrounded by mixed anions, fluorine and oxygen ions. The decrease of Ω_2 Judd–Ofelt parameters with increasing fluorine content suggest this increase of the number of fluorine ions surrounding erbium ions. Moreover, the introduction of fluorine for a constant ratio of Zn/Te leads to a slight depolymerization of the glass network presenting several tellurium groups which can also contribute in minor proportion to change the local crystal field. This can explain the increase of the linewidths for ETZF-1 and ETZF-2 resulting in a higher variation in the environment of Er^{3+} from one site to another site. On the other hand, a lesser site-to-site variation can be expected in the case of ETZF-5 where the dimensionality and the connectivity of the glass decrease with increasing the proportion of TeO_3 and TeO_{3+1} groups as it was shown by Raman spectroscopy and XR diffraction analysis. The thinning of the emission peak observed may be related to the homogeneity of the shell-coordination of the erbium ions by increasing the fluorine contents. The emission efficiency from $^4I_{13/2}$ to $^4I_{15/2}$ levels depends highly on the $^4I_{11/2}$ level lifetime. The increase of $^4I_{11/2}$ lifetime influenced by the introduction of fluorine is due to radiative decay rate change and a likely decrease of multiphonon process. Even if the phonon energy seems not to change according to Raman spectroscopy, the multiphonon relaxation rate of the $^4I_{11/2}$ to the level $^4I_{13/2}$ can be affected by the introduction of fluorine. The Raman spectroscopy only provides an averaged vibrational density and the behavior of the phonon mode locally coupled to the rare-earth ion in the glass could be different since an electron–phonon coupling strength decreases has been reported as probable in oxyfluoride tellurite glasses [41]. Its contribution to decrease the emission efficiency from $^4I_{13/2}$ level might be taken into account for the fluorine richest glasses and an introduction of cerium ion should improve the 980 nm excitation efficiency for this composition [43].

The up-conversion fluorescence efficiency is correlated to the probability of multi-step excitation by the excited-state absorption (ESA) or the energy transfer between adjacent excited ions, and the quantum efficiency of the emitting level. The probability of these processes is related to the lifetime of the intermediate excited level, $^4I_{11/2}$ and $^4I_{13/2}$. Indeed, the increase of upconversion fluorescence follows the same trend as lifetime decay rate observed for the $^4I_{11/2}$ and $^4I_{13/2}$ excited levels which increases with the growing introduction of fluorine and/or the decrease of TeO_2 contents. The green luminescence results from absorption of a second photon at 974 nm from the $^4I_{11/2}$ excited state. The red luminescence may have two different origins. The $^4F_{9/2}$ level can be populated by multiphonon relaxation down from $^2H_{11/2}$, $^4S_{3/2}$ level initially excited by absorption of a second photon from the $^4I_{11/2}$ excited state. The $^4I_{13/2}$ excited state can also contribute to populate the levels $^4F_{9/2}$ by absorption of a second photon (Fig. 3). The increase of the red to green intensity ratio is in accordance with an up conversion process dominated by absorption from the $^4I_{13/2}$ excited level in the case of oxyfluoride tellurite glasses induced by an increasing lifetime of the $^4I_{13/2}$.

5. Conclusions

In the present investigation, the glass forming capability, glass network structural organization and optical properties have been examined in the rare earth-doped oxyfluoride zinc tellurite glass system. In the range of 75–47 mol% of TeO_2, the thermal stability of the glass with respect to crystallization was slightly improved by incorporating ZnF_2 in the glass system. Addition of fluorine results in reduction of Te–O–Te linkages due to a gradual transformation of trigonal bipyramid TeO_4 through TeO_{3+1} to trigonal pyramid TeO_3 decreasing the connectivity of the tellurium former network. This behavior is strengthened by the higher possible ratio $(ZnO–ZnF_2)/TeO_2$ compared with oxide tellurite glasses favored by fluorine substitution. For a constant ratio of Te/Zn, the effective bandwidth of absorption cross-section and normalized emission increases while fluorine concentration is growing. This broadening of the emission peak should be related to the changes of the host glass structure and the surrounding coordination of the erbium ions due to the presence of mixed anions, fluorine and oxygen. Three different types of structural units in the former network of tellurite glass involve an important site-to-site variation of the ligand field favored by the introduction of fluorine in controlled proportion. The particular feature of the fluorine richest tellurite glass compared to the other oxyfluoride should be explained by a different structural organization of the former network surrounding the Er^{3+} mainly constituted of entities more close as those found in $ZnTeO_3$ or $Zn_2Te_3O_8$. Additionally, the Er^{3+} lifetime of the emitting level $^4I_{13/2}$ increases drastically with increasing fluorine content due mainly to the radiative rate change and by eliminating non-radiative decay rates related to the water vibrational frequency. The co-doping of cerium ion to the Er-doped low-phonon energy glasses should improve the 980nm excitation efficiency by depopulating the level $^4I_{11/2}$ and will may be the next focus of investigation in oxyfluoride tellurite glasses.

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